

Notes Week 25

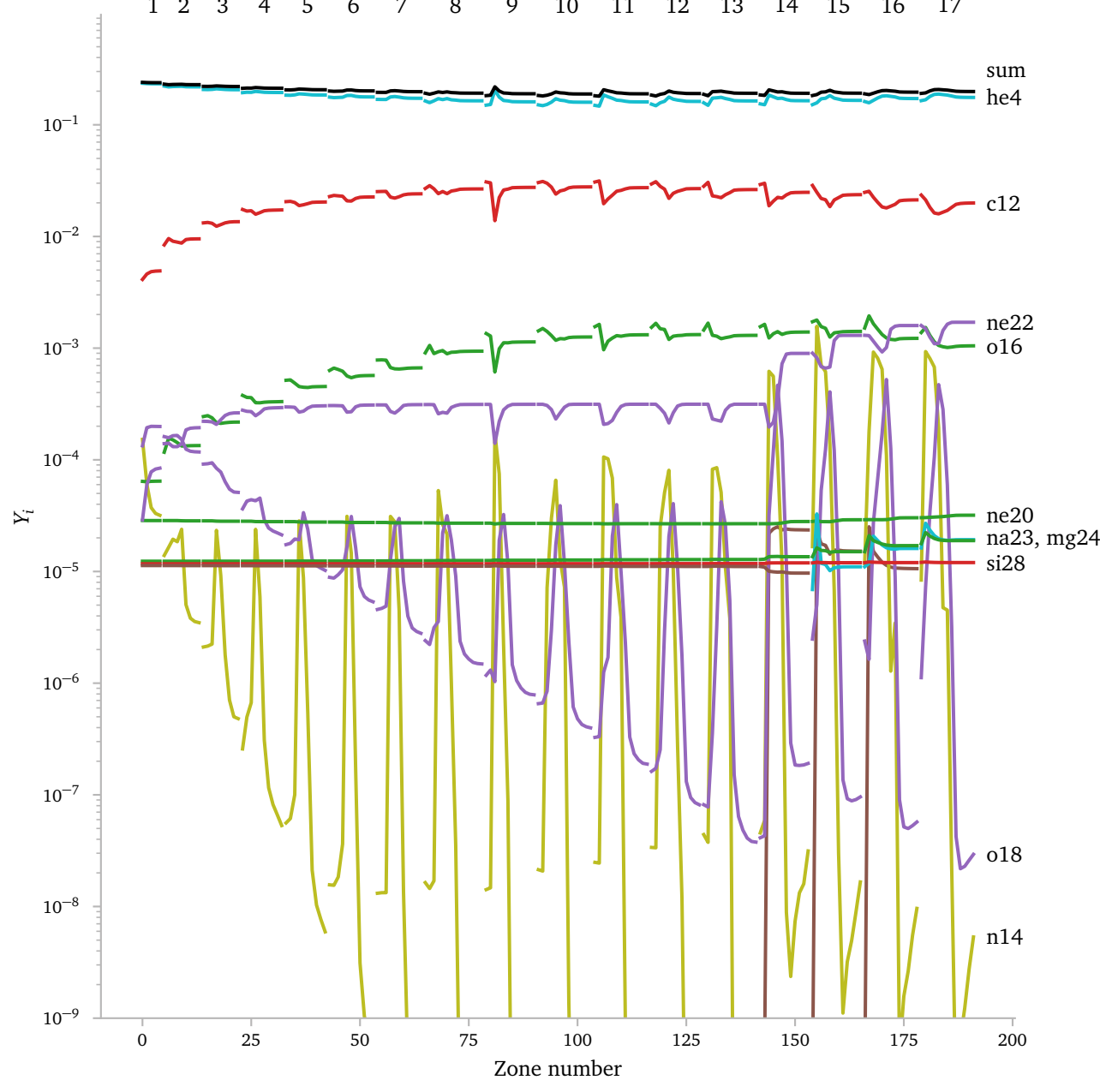
a. Intershell abundances

Using the provided script I was able to, quite easily access the intershell abundances.

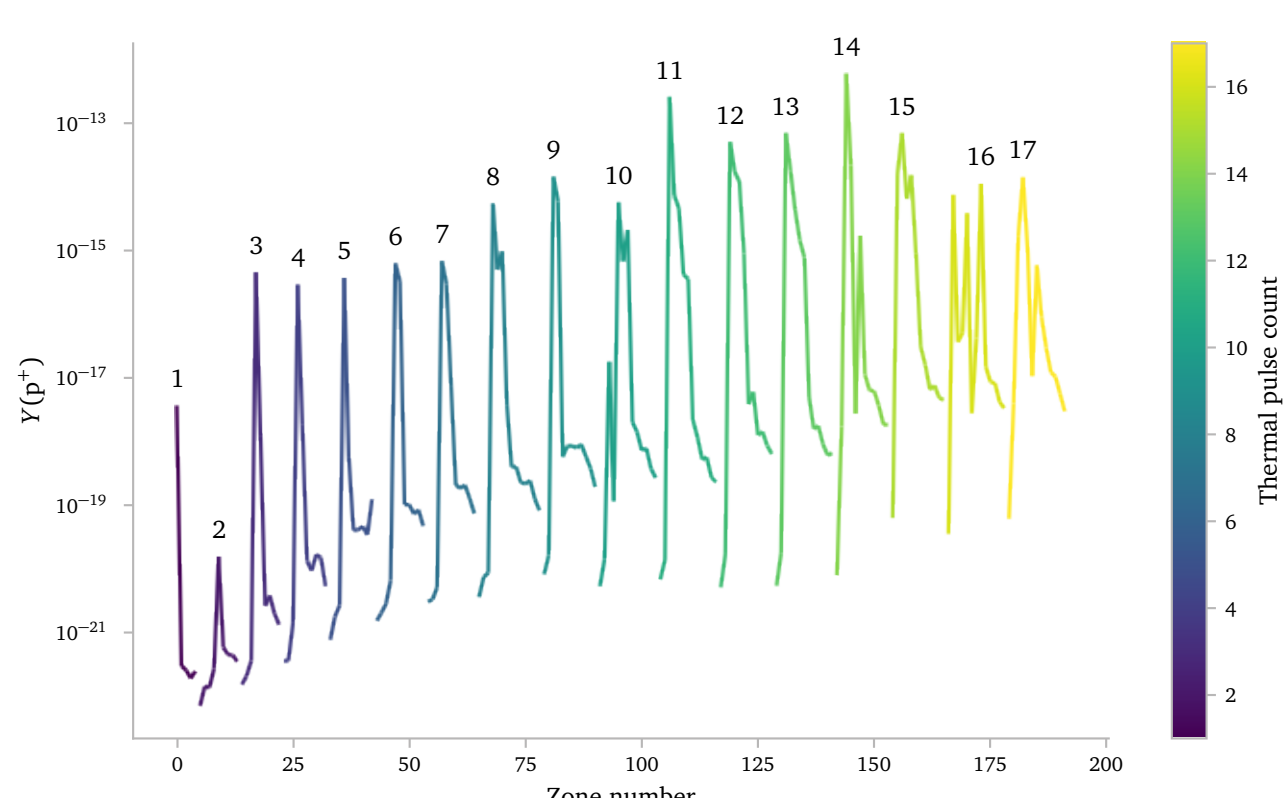
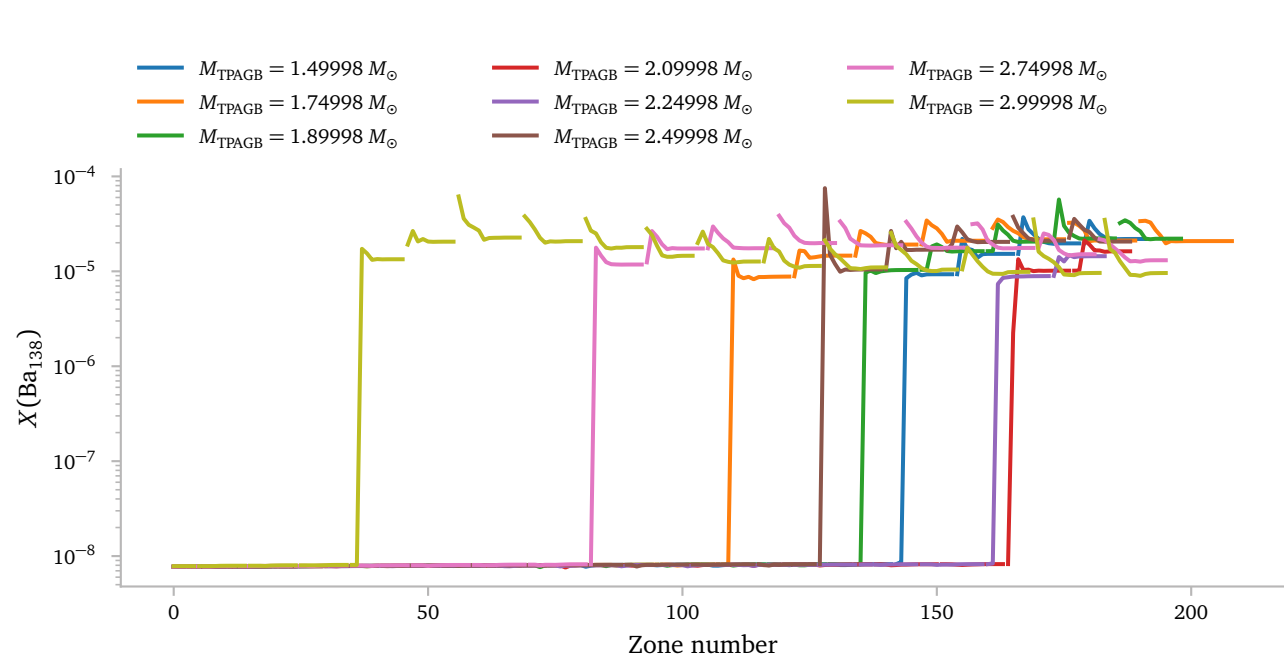
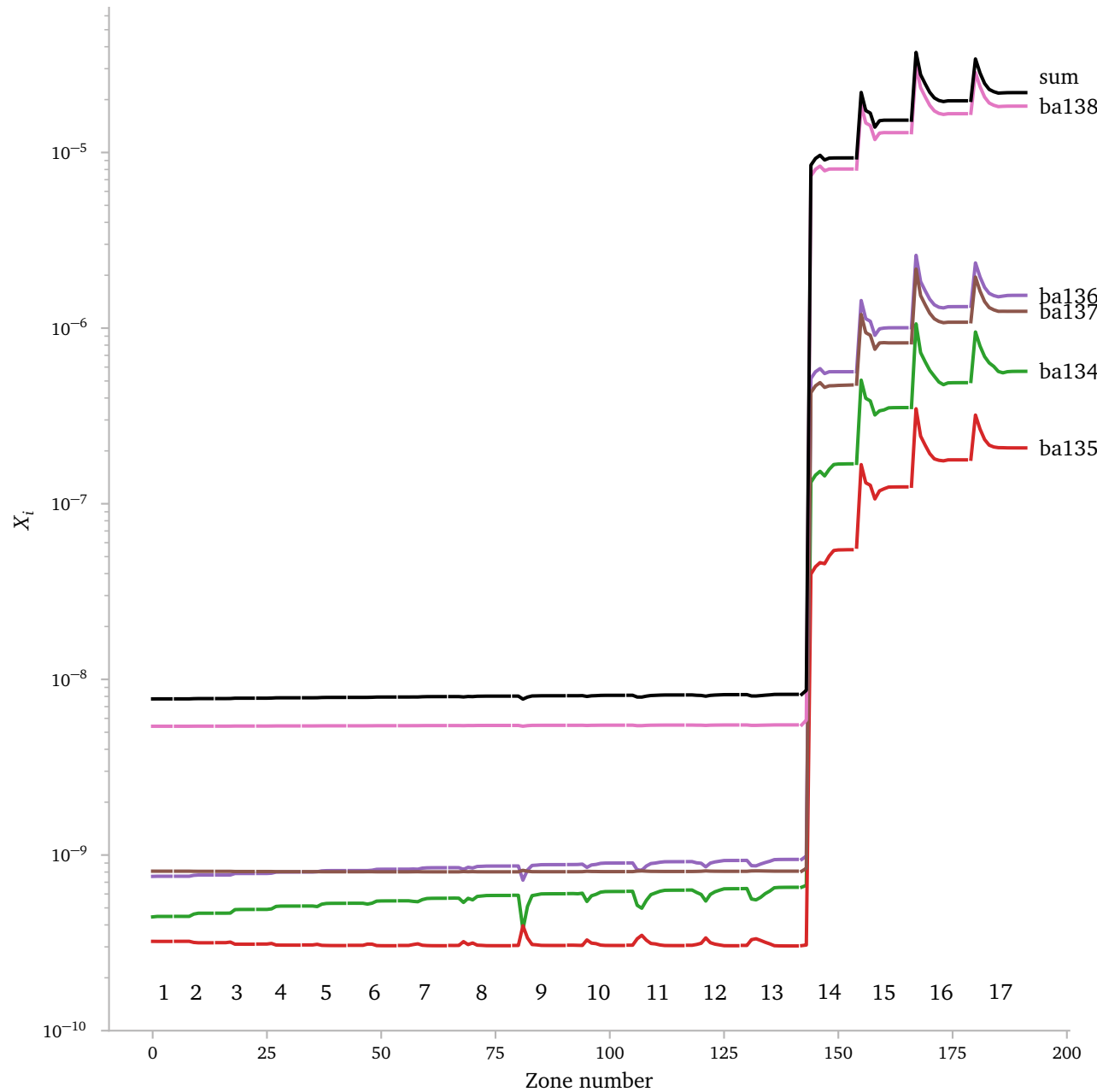
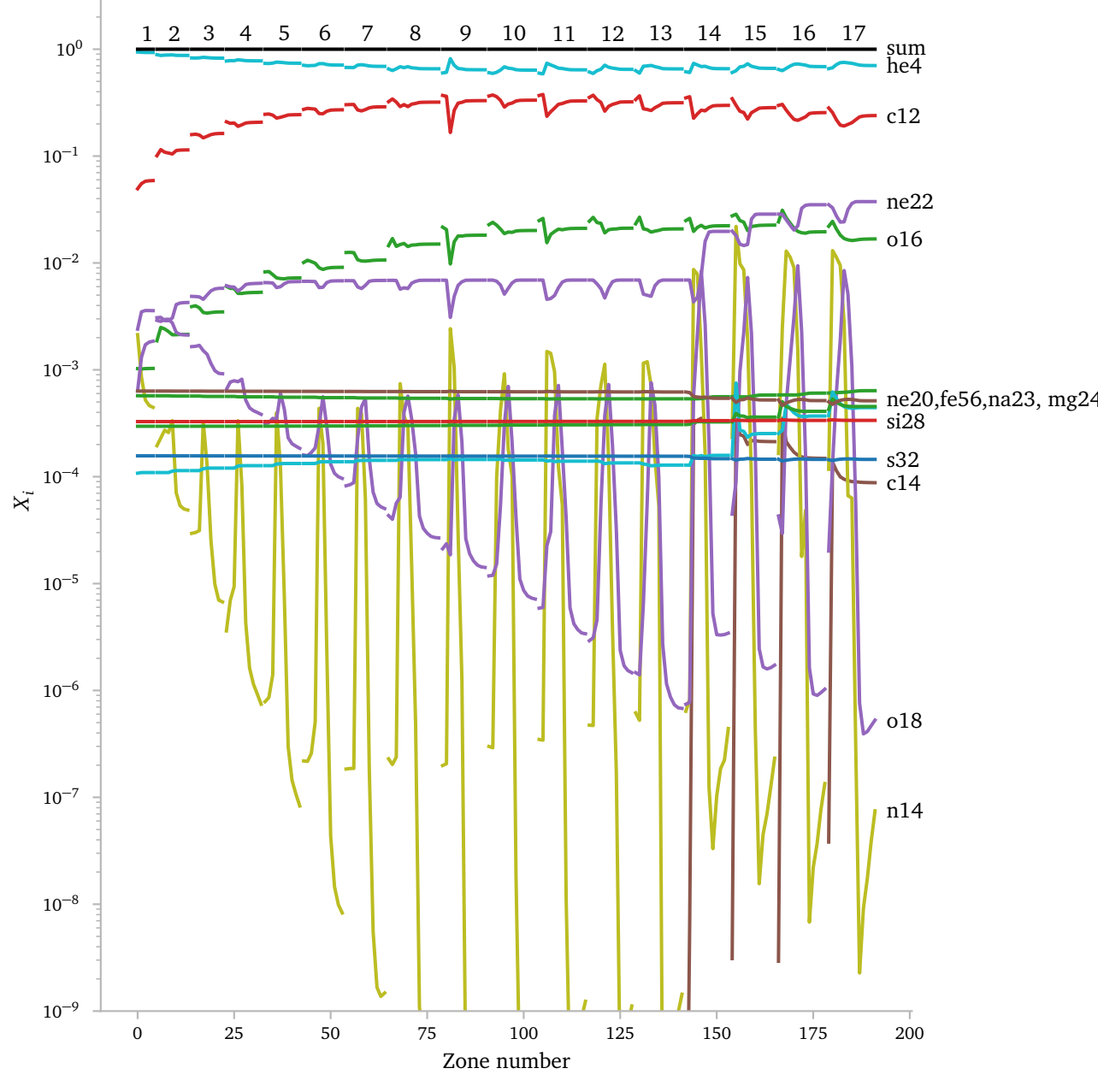
However, the structure is not immediately very clear to me. It seems like there are multiple abundances for every.

Then I also need to figure out what the actual “abundances” mean.

A quick test is to simply sum the “abundances” of a TP and see what the sum is equal to.



Clearly, this is *not* a mass ratio, since this would sum to 1. Now, if we multiply it by the atomic mass, we get the following picture.



b. Fixing MESA models

Overall, my code to “revive” MESA models works very well!

```
cleaning dir
making binaries
11800, not enough models between
11750, possible, testing retries
retries succesful
x750
18250, not enough models between
18200, possible, testing retries
retries succesful
x200
18200, not enough models between
18150, possible, testing retries
retries succesful
x150
18250, not enough models between
18200, possible, testing retries
retries succesful
x200
18300, not enough models between
18250, possible, testing retries
retries failed
18200, want to select the same starting model as last
-- time. selecting earlier model.
18150, possible, testing retries
retries succesful
x150
18250, not enough models between
18200, possible, testing retries
retries succesful
x200
18300, not enough models between
18250, possible, testing retries
retries failed
18200, want to select the same starting model as last
-- time. selecting earlier model.
18150, possible, testing retries
retries succesful
x150
18250, not enough models between
18200, possible, testing retries
retries succesful
x200
18300, not enough models between
18250, possible, testing retries
retries failed
18200, want to select the same starting model as last
-- time. selecting earlier model.
18150, possible, testing retries
retries succesful
x150
18250, not enough models between
18200, possible, testing retries
retries succesful
x200
30050, not enough models between
30000, possible, testing retries
retries succesful
30000
30000, not enough models between
29950, possible, testing retries
retries succesful
x950
30050, possible, testing retries
retries failed
30000, possible, testing retries
retries succesful
30000
30100, not enough models between
30050, possible, testing retries
retries failed
30000, want to select the same starting model as last
-- time. selecting earlier model.
29950, possible, testing retries
retries succesful
x950
30050, possible, testing retries
retries failed
30000, possible, testing retries
retries succesful
30000
30100, not enough models between
30050, possible, testing retries
retries failed
30000, want to select the same starting model as last
-- time. selecting earlier model.
29950, possible, testing retries
retries succesful
x950
30050, possible, testing retries
retries failed
30000, possible, testing retries
retries succesful
30000
30100, not enough models between
30050, possible, testing retries
retries failed
30000, want to select the same starting model as last
-- time. selecting earlier model.
29950, possible, testing retries
retries succesful
x950
```

c. Interesting model

R500_00_q0.700_eps0.250_delta0.200_M3.0/ resume-past-instability.py
This model has a very late thermal pulse.

References